

Model based decision systems fail silently

Every component metric below improved, or looked correct, while the decision degraded or the constraint was violated. Each metric is computed inside the model's own account of the world. The right hand column is the measurement that exposed the gap.

THE COMPONENT LOOKED CORRECT BY	BUT IN FACT	THE MEASUREMENT THAT EXPOSED IT
surrogate episode cost, dual satisfied	7.3 percent of evaluations violated the true limit	probe the true environment periodically
one step forecast skill	open loop rollout invents a fire that does not exist	synchronise to observation, compare to open loop
faithfulness 0.47 against 0.18, t of 21	briefs carry zero state specific information	permutation control, score against another state
cost separates between doing nothing and acting greedily	methods were ranked inside a 7 percent achievable band	return separation, not only cost separation
four constrained RL baselines appeared to fail structurally	they had barely moved off their initialisation	start every baseline where the method starts
reconstruction precision 0.594 against 0.006, about 100 times better	the decision was worse in 0 of 9 configurations	score the belief on the decision it produces
a margin cut violations about 7 times	no stated failure rate, and the first conformal attempt made things worse	conformalise the distribution that actually violates

The positive results share one shape: measure the quantity that actually matters, which is real cost, observed state, or decision outcome, and the system works.